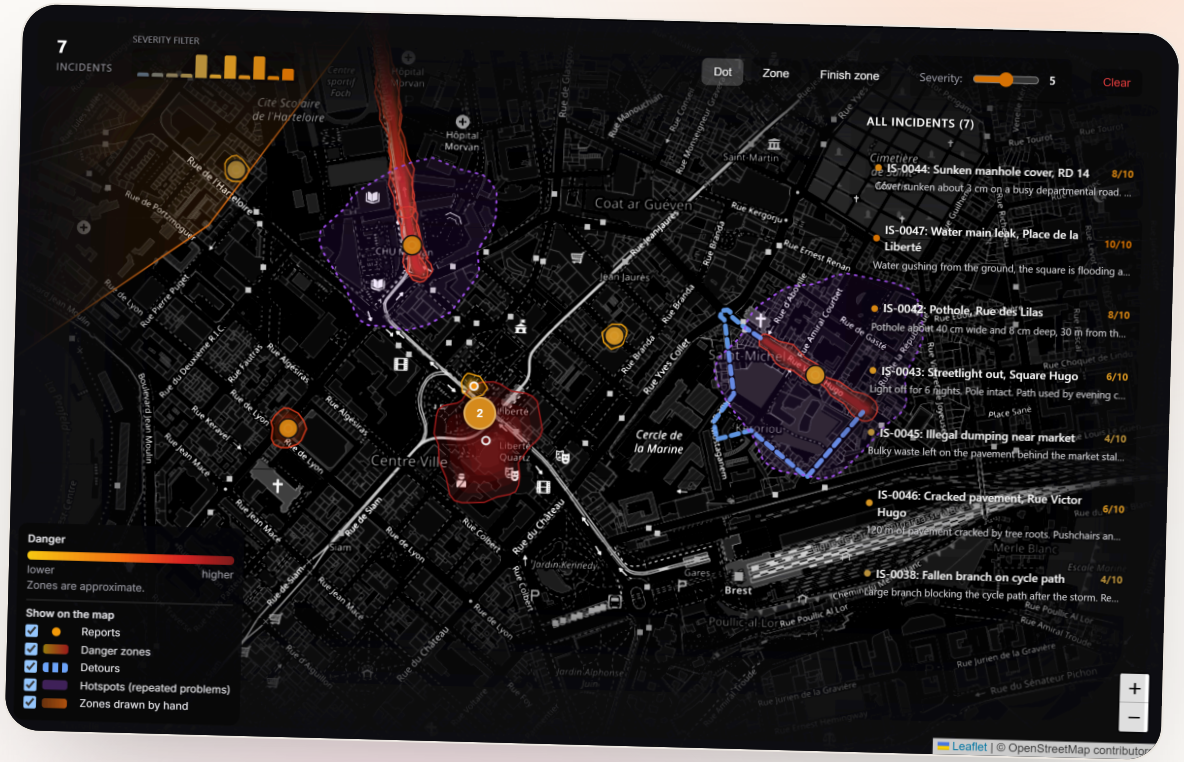
InfraSight

The AI co-pilot for city streets.

A photo from a citizen becomes a mapped danger, a merged case, a detour, a draft order and a message back home. Built on Meta's models.

- Muse Spark
- SAM 3.1
- Muse Voice Transcribe
- OpenStreetMap



THE PROBLEM

Every pothole is a small administrative maze.

A town hall can get hundreds of reports a year. Each one has to be understood, checked, routed, approved, bought, signed and closed, while citizens wait without news.



Unclear reports

Blurry photos, vague addresses, the same hole reported five times.



Who is responsible?

Commune, département, State, intercommunality, network operator or private owner.



Rules at every step

Purchase thresholds, council votes, DT-DICT, traffic orders, acceptance reports.



Silence for citizens

People rarely hear back, so they report again, or stop reporting.

BUILT ON META AI

Three Meta models, one system.



Muse Voice Transcribe

Next

→ Citizens report by speaking

→ Any language, hands-free

→ Text goes straight to triage

→



Muse Spark

Built

→ Reads the photo and the text: triage

→ Shapes the danger zone, spots duplicates

→ Drafts reports, orders and stage documents

→ Compares before / after photos

→ Writes updates in 9 languages, weekly brief, forecast

↔



SAM 3.1

Server ready · GPU

→ Outlines the hole or crack on the photo

→ The outline feeds the size and cost estimate

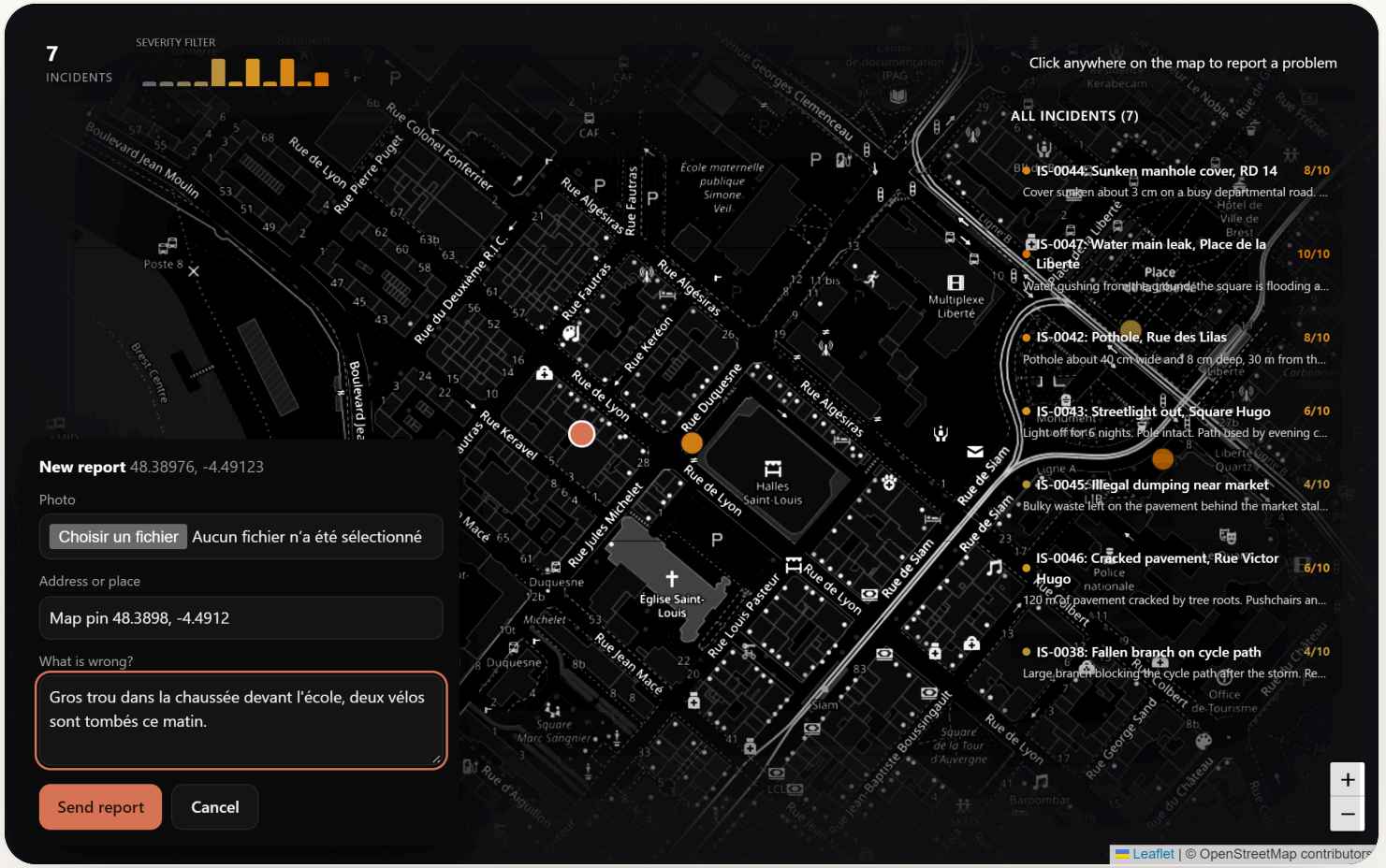
→ Shown in red over the citizen's photo

Grounded in real-world data: addresses, street shapes and the road network come from OpenStreetMap, not from the model's memory.

FOR CITIZENS

Report in 30 seconds.

- Tap the spot on the map, the address fills itself.
- Add a photo and a few words, in your own language.
- Muse Spark classifies it and finds who is responsible.
- Updates arrive in the language you chose.

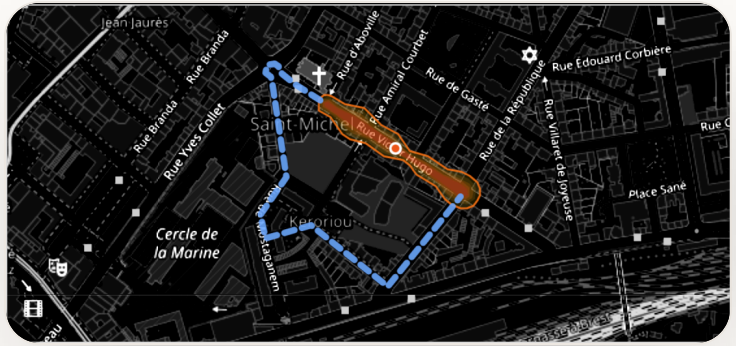


Screenshot of the working prototype, demo data in Brest

MUSE SPARK × OPENSTREETMAP

Danger zones shaped by the problem.

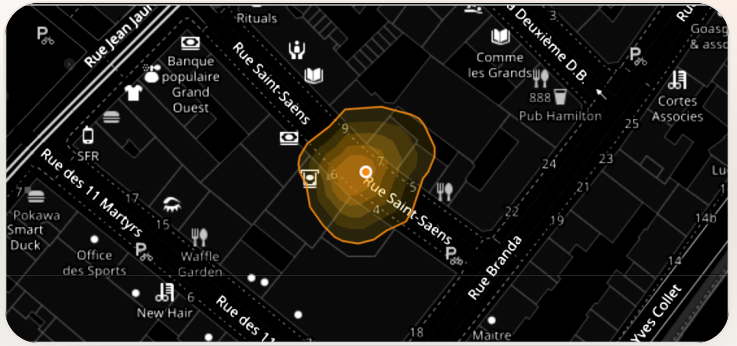
Muse Spark decides what kind of danger it is. InfraSight draws it on the real street: from yellow when the danger is low to red when it is severe or the road is blocked.



Blocked pavement · drawn along the whole street section



Water main leak · spreads over the square



Streetlight out · a small local zone

Danger — lower → higher

Zones are approximate: they guide action, they are not measurements. Each layer (reports, zones, detours, hotspots) can be shown or hidden on the map.

MUSE SPARK

Five reports, one case.

When a new report lands near an open case, Muse Spark is given the nearest cases with their distance and decides if it is the same problem. If so, the report joins the case: its photo goes in the history and the citizen follows the case.



Less work

One case to process instead of many.

Severity can rise

More reports and photos make a stronger case.

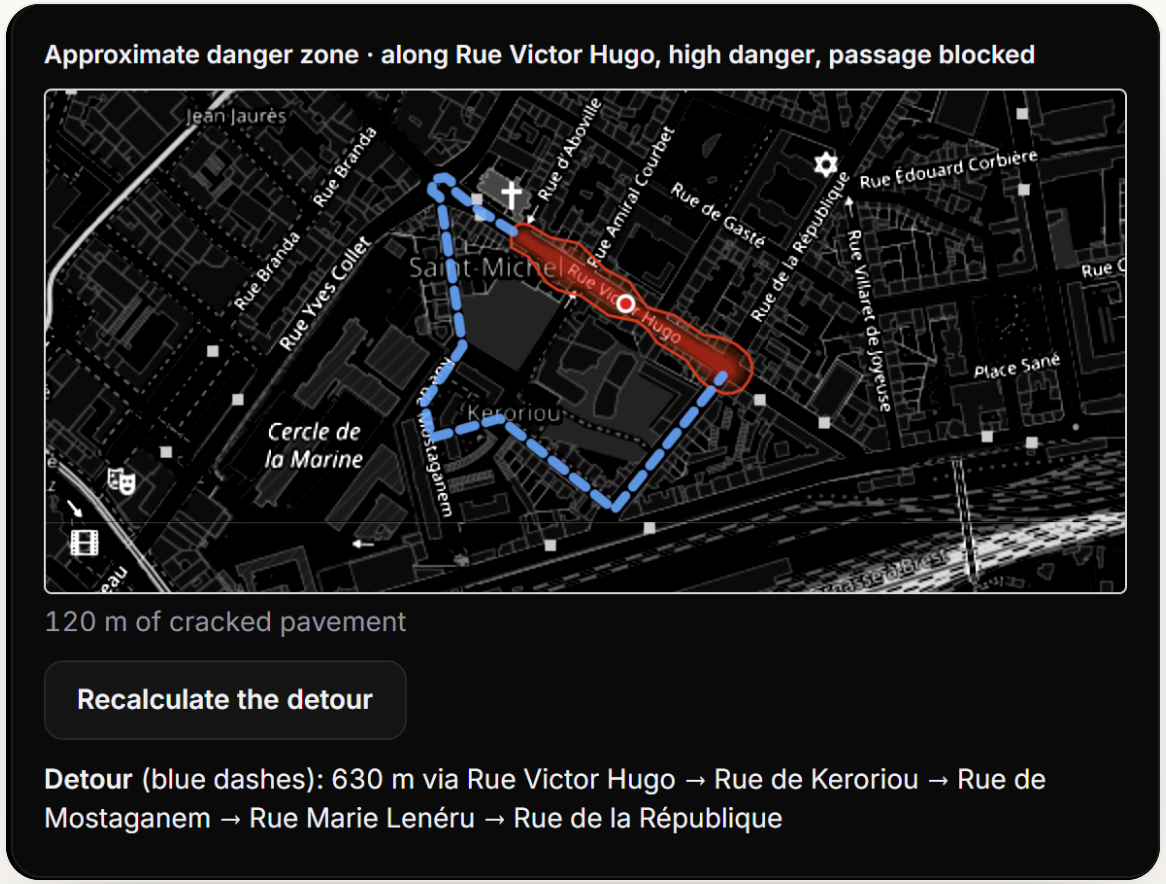
Everyone is told

All citizens who reported it get the same updates.

Screenshot of the working prototype

REAL DATA, REAL STREETS

The road is blocked. Here is the way around.



- InfraSight downloads the road network around the blocked section from OpenStreetMap.
- It computes the shortest legal detour, one-way streets included: here 630 m.
- Muse Spark drafts the mayor's traffic order in French, with the detour streets in it.
- Staff check it; the mayor signs.

Why it matters

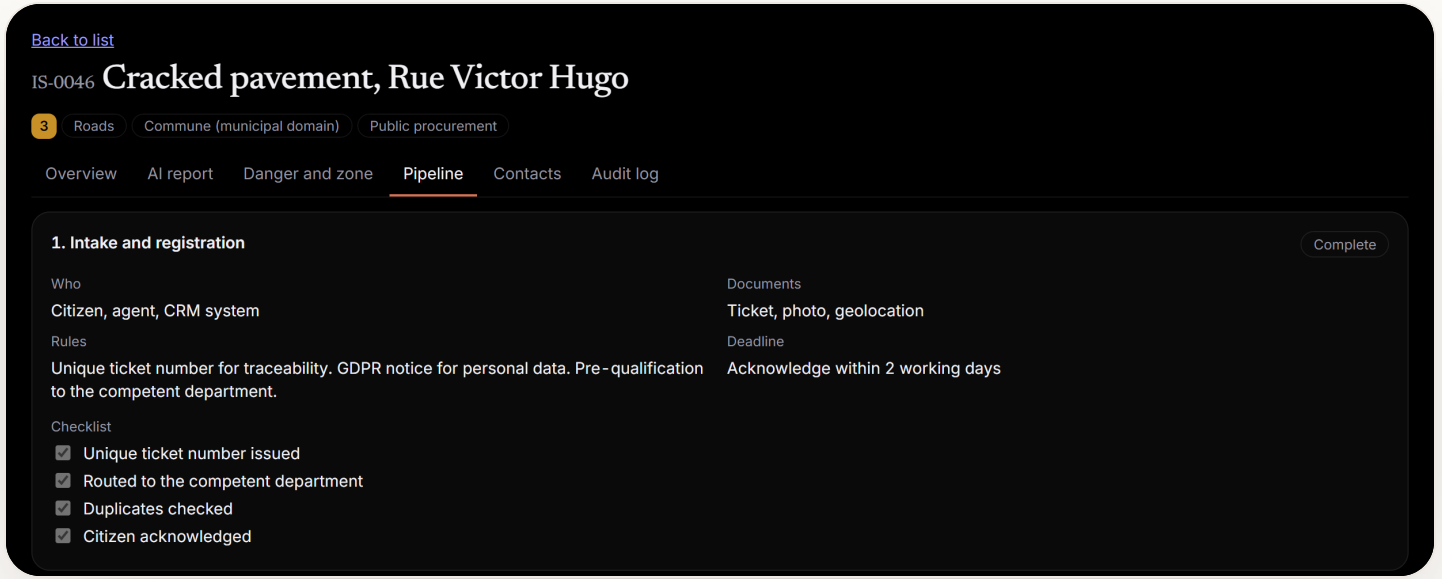
A traffic order normally takes hours of manual work: finding the detour, writing the articles, checking the legal basis.

Screenshot of the working prototype

MUSE SPARK FOR STAFF

The paperwork, drafted for you.

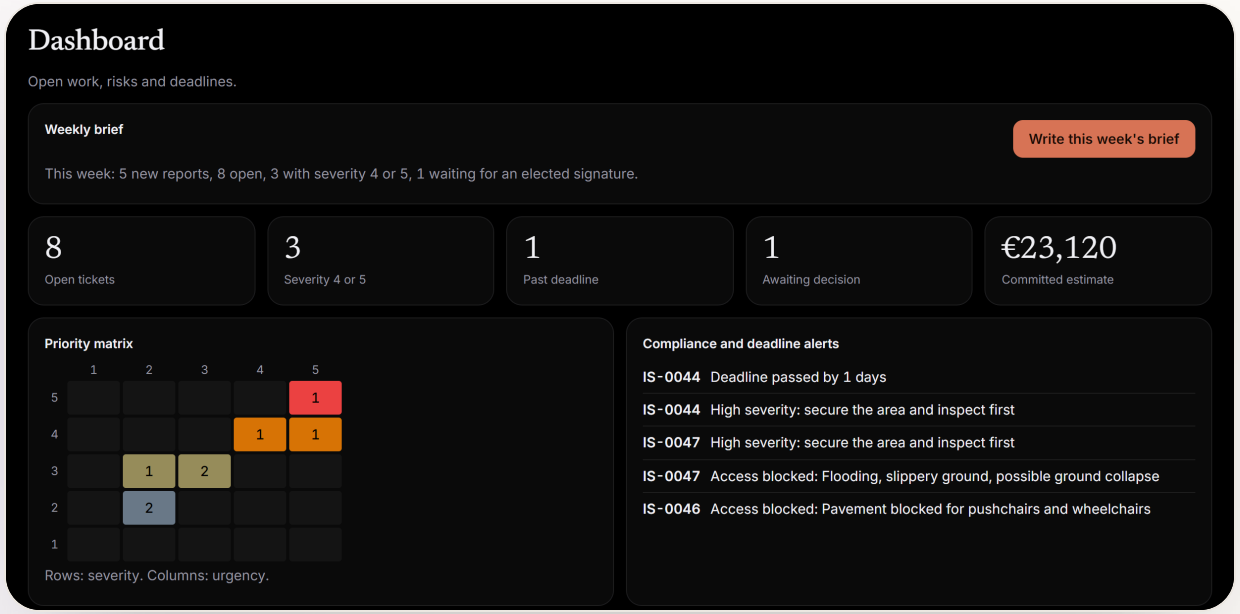
- 7 legal stages, from intake to acceptance, with the rules and deadlines of each.
- "AI assist" drafts the document of the current stage: decision memo, specifications, permits checklist, acceptance report.
- At closing, Muse Spark compares the before and after photos.
- Only an elected official can sign the decision stage.



Screenshot of the working prototype

FOR ELECTED OFFICIALS

See the whole city, act before it breaks.



Dashboard: priorities, alerts, and a weekly brief written by Muse Spark from the case data.

Repeated problems

Street	Reports	Open	Types	Avg sev.	Cost	Last
Avenue du Maréchal Foch	3	1	Roads 2, Water 1	3.7	€6,500	2026-11-01
Rue Victor Hugo	4	1	Roads 4	2.8	€25,850	2026-10-17

Hotspots: streets where problems keep coming back. Muse Spark turns them into a forecast with preventive actions and a budget.

WHY IT IS NOT "JUST A CHATBOT"

AI that works on the real world.

Multimodal

It reads photos, not only text: triage, duplicates, before / after checks. SAM 3.1 adds pixel-level outlines.

Grounded in real data

Real addresses, street shapes and road network from OpenStreetMap. The legal rules of each stage are given to the AI, not left to its memory.

It produces actions

Zones on the map, merged cases, detours, draft orders, citizen messages. Not just answers in a chat window.

Humans decide

Every AI output is labelled in the history. Staff correct it; only an elected official signs.

For everyone

Updates in 9 languages, including Breton and Arabic, each citizen in their own.

Robust

Accounts and roles, prompts on the server, usage limits. If the AI is down, reports are still saved.

DURING THE HACKATHON

From co-pilot to agent.

Next

InfraSight Agent

Muse Spark with tool calling handles a report end to end: looks at the photo with SAM 3.1, reads the road data, finds nearby cases, estimates the cost, checks the purchasing rules, plans the detour. Every step visible live; every change waits for a staff click.

Next

Voice reports

Muse Voice Transcribe: a citizen speaks, the agent takes it from there. Useful for people who can't type easily, and on the go.

Next

Learns from staff

Each correction by a technician is stored and reused. The dashboard shows how often the AI was right, and sends uncertain cases straight to a human.

Ready SAM 3.1 server written and wired in: it runs on the hackathon GPU and plugs into the damage outline and the agent.

WHERE WE ARE

A working prototype, ready for a pilot.

- Citizen, technician and elected-official accounts
- Muse Spark triage, reports, drafts, updates, forecast, brief
- Danger zones, duplicate merging, detours, hotspots
- SAM 3.1 damage outline server, ready on GPU

AFTER THE HACKATHON

A 3-month pilot with one commune

- Real reports from one neighbourhood
- Legal review of every rule built into the stages
- Measure response times and citizen satisfaction

Let's fix streets faster, together.